

Grade 8
Unit 4
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Write an exponential expression equivalent to 8^3 that has a base of 2.

$$(2^3)^3 = 2^9$$

2. Write an exponential expression equivalent to 25^3 that has a base of 5.

$$(5^2)^3 = 5^6$$

3. Write an exponential expression equivalent to 64^2 that has a base of 8.

$$(8^2)^2 = 8^4$$

Marquise says the expressions $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$ and $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$ are equivalent because both expressions can be rewritten as 2^{-12} .

In questions 4–5, generate an equivalent expression for each expression using a base of 2. Show all work.

4. $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$

$$\left(\frac{1}{8}\right)^3 \cdot \frac{1}{2^3} \Rightarrow \left(\frac{1}{2^3}\right)^3 \cdot \frac{1}{2^3} \Rightarrow \frac{1^3}{2^9} \cdot \frac{1}{2^3} \Rightarrow \frac{1}{2^{12}} \Rightarrow 2^{-12} \checkmark$$

5. $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$

$$\left(\frac{1}{4}\right)^5 \cdot \frac{1}{2^2} \Rightarrow \left(\frac{1}{2^2}\right)^5 \cdot \frac{1}{2^2} \Rightarrow \frac{1^5}{2^{10}} \cdot \frac{1}{2^2} \Rightarrow \frac{1}{2^{12}} \Rightarrow 2^{-12} \checkmark$$

6. Is Marquise correct? Explain your answer.

Yes, Marquise can use the negative exponent property, the power of a quotient property, the power of property, and the product property to rewrite questions 4 and 5 as equivalent expressions.

In each column, generate an equivalent expression using as few bases as possible.
Then match Column A expressions with expressions in Column B that are equivalent.

Column A	Work	Column B	Work
7. $2^3 \cdot 2^4$	2^{3+4} 2^7	A. $3^{-3} \cdot 2^6$	$\frac{1}{3^3} \cdot \frac{2^6}{1}$ $\frac{2^6}{3^3}$
8. $\frac{2^6}{3^3}$ Does not simplify	$\frac{2^6}{3^3}$	B. $\frac{(3 \cdot 2)^{-2}}{3^{-4}}$	$\frac{3^4}{(3 \cdot 2)^{+2}} \cdot \frac{3^4}{3^2 \cdot 2^2}$ $\frac{3^2}{2^2}$
9. 2^0	Anything to 0 power = 1	C. $\left(\frac{1}{2}\right)^{-3} \cdot \left(\frac{1}{2}\right)^{-4}$	$\frac{1^{-3}}{2^{-3}} \cdot \frac{1^{-4}}{2^{-4}}$ $\frac{2^3}{1^3} \cdot \frac{2^4}{1^4} = \frac{2^{3+4}}{1} = 2^7$
10. $\left(\frac{3}{2}\right)^2$	$\frac{3^2}{2^2}$	D. 1 Does not simplify	1

Question 7 in Column A is equivalent to Letter C in Column B.

Question 8 in Column A is equivalent to Letter A in Column B.

Question 9 in Column A is equivalent to Letter D in Column B.

Question 10 in Column A is equivalent to Letter B in Column B.